

Designing a Minecraft World That Carries Identity and Community Knowledge

Mini-Unit Packet

Layered Genius: Designing for Joy and Identity

Grade 4–5 Computer Science / Digital Literacy mini-unit centered on Minecraft Education, multimodal meaning-making, identity-safe design, and community knowledge

Student	Piter Zacari Garcia Bautista
Course	EDU498: Literacy Learning as Social Practice
Class Session	Class 7: Towards a Pursuit of Identity
Assignment	Layered Genius: Designing for Joy and Identity
Mini-Unit Focus	Minecraft world-building as identity, community, and coding literacy
Student Profile	Ethan, Student 5, Computer Science
Submission Date	June 10, 2026

This draft uses my Puzzle Plan and EQUITAS lenses to treat Minecraft Education as a literacy space where students can build, code, debug, explain, and represent meaning without being reduced to a single narrow academic mode.

Note: *This work focuses on Joy and Identity, while adding each piece visible in stronger mini-unit examples: a fuller lesson sequence, all five pursuits, assessment evidence, a sample model, and student-facing appendix tools.*

Mini-Unit Snapshot

Task: integrate Joy + Identity into my previous unit to create a mini-unit packet that also names Skills, Intellect, and Criticality clearly enough for feedback and future expansion.

Grade / discipline: Grade 4–5 Computer Science / Digital Literacy

Student profile: Ethan, a Seneca / Haudenosaunee and mixed-identity student who enjoys coding, game design, robotics, collaboration, outdoor learning, and community storytelling.

Mini-unit focus: Students use Minecraft Education to plan and begin building a small digital place that communicates identity, community knowledge, belonging, or future possibility.

Essential question: How can code, design, and digital place-making communicate who we are, what matters to us, and what our communities know?

Unit context and purpose: This three-lesson unit is designed for a Grade 4–5 Computer Science / Digital Literacy setting using Minecraft Education. Students use digital world-building as a literacy practice: they plan, code, debug, label, explain, and share a digital place that carries meaning. The purpose is to help students see coding as authorship, identity, community knowledge, and audience-aware design.

The unit foregrounds Joy and Identity by giving students choice, visible progress, and a protected way to connect design with who they are or what they value. Skills, Intellect, and Criticality are named as developing layers so the mini-unit can grow without losing its central focus on identity-safe participation.

Three-lesson Arc:

- **Lesson 1: Plan a Meaningful Place.** Choose a place concept, sketch it, name meaningful features, and begin one prototype.
- **Lesson 2: Build and Debug.** Use Minecraft / MakeCode to build, test, revise, and add at least one coded or repeated design feature.
- **Lesson 3: Share and Reflect.** Present the world, explain design choices, and respond to peer feedback.

Source Trail and Rationale for the Pick

This work grows from my work toward “*Layered Genius: Designing for Joy and Identity*” and from Muhammad’s Identity pursuit, which frames identity as self-definition. It also carries forward my joy commitment of “belonging before building” and my placement observations of debugging, collaborative work, peer teaching, and student-led problem framing. I also draw on Moje, Luke, Davies, and Street’s literacy-and-identity lens, Skerrett’s attention to repositioning students as capable contributors, and Moje’s disciplinary literacy framework. These sources help name coding, debugging, design, and explanation as literacy practices. Minecraft Education/MakeCode and the CSTA/ISTE standards serve as practical anchors for sequencing, debugging, computational thinking, creative communication, collaboration, and multimodal digital design.

Student Profile and Identity Safety

Ethan's profile matters because the lesson can recognize him as a coder, designer, storyteller, and community-minded problem solver. At the same time, the lesson should honor his identity without requiring him to explain, display, or represent his culture for the class. Instead, it offers identity-safe choice by allowing students to build from lived experience, imagination, community, interests, family, land, language, hobbies, or future hopes. Because Ethan's profile includes Seneca / Haudenosaunee heritage, culturally specific symbols or stories should be student-chosen and handled with care. Private disclosure is never required. Fictionalized, symbolic, or aspirational places are valid.

Identity design move: The lesson treats identity as a design resource, not a disclosure requirement. Students choose what counts as meaningful, and the teacher assesses how clearly design choices communicate meaning. This keeps the work aligned with our focus on identity while respecting privacy, agency, and student control.

Pursuits Alignment

Pursuit	How to Address it
Joy	Students experience choice, visible progress, peer support, debugging as discovery, and the satisfaction of building something shareable.
Identity	Students design a meaningful place and explain how specific features communicate belonging, community, memory, culture, or future possibility.
Skills	Students practice sequencing, decomposition, loops or repeated patterns, debugging, spatial reasoning, screenshot use, partner communication, and oral/written explanation.
Intellect	Students analyze how code, space, signs, paths, color, interaction, and audience work together to create meaning in a digital world.
Criticality	Students ask whose stories do/don't appear, how accessibility shapes belonging, and what respectful use means when cultural symbols or community knowledge are involved.

Multimodal and Multiliteracies Design

Minecraft functions here as a text students author. They read digital space, sketch a plan, build visually, use procedural thinking, explain orally, label in writing, and revise through peer feedback.

Modes included: spatial design, visual design, coding/sequencing, oral explanation, written labels or reflection, screenshots, collaboration, and peer feedback.

Access moves: sketch-first or build-first choice; projected model; partner roles; sentence stems; optional oral explanation; screenshots as memory support; chunked directions; low-floor/high-ceiling challenge options.

Why this supports multiliteracies: Students are not only writing or answering questions; they are composing meaning across space, image, code, talk, labels, screenshots, and peer feedback. A path, sign, repeated pattern, gathering space, or coded sequence can all function as evidence of meaning-making.

Lesson 1 Plan

Duration: 60 minutes

Lesson title: Planning and Prototyping a Meaningful Minecraft Place

Student-friendly target: I can plan and begin building a Minecraft place that shows something important about identity or community, and I can explain how my design choices carry meaning.

Identity-safe teacher language: Your place can be real, imagined, symbolic, collaborative, or future-facing. You do not have to share anything private. The goal is to make thoughtful design choices and explain what those choices communicate.

Objectives:

1. Students will choose a meaningful place connected to identity, community, belonging, or future possibility.
2. Students will break the place into smaller build/code steps.
3. Students will begin one prototype and test or revise at least one feature.
4. Students will explain one design choice using meaning, audience, and purpose.

Materials: Minecraft Education, teacher model world, projector, planning sheet, pencils or markers, optional MakeCode reference cards, and exit slip. If devices fail, students complete a paper-world storyboard and mark which steps would later be coded.

Standards connection: CSTA 1B-AP-10, 1B-AP-11, 1B-AP-15, and 1B-AP-17; ISTE Innovative Designer, Computational Thinker, Creative Communicator, and Global Collaborator.

Mins	Activity	Purpose
0–7	Show a generic Minecraft structure and a meaningful place. Ask: “What makes a place meaningful, not just well built?”	Frames design as meaning-making.
7–15	Model a small place plan: entrance, sign, path, gathering spot, and one repeated pattern that could be built or coded.	Makes decomposition and design choices visible.
15–25	Students choose a real, imagined, symbolic, or future place and sketch 3–5 important features.	Builds identity-safe agency.
25–38	Students label 3–5 build/code steps and begin a prototype in Minecraft or on paper.	Moves from idea to manageable action.
38–50	Students test one feature, debug one issue, or revise one design choice.	Treats mistakes as part of learning.
50–57	Partner share: “This place matters because...” and “One choice I made was...”	Centers oral literacy and audience.
57–60	Exit slip: “What does your place show? What is your next build/code step?”	Captures evidence and next steps.

Differentiation: Students may work alone or with roles such as builder, navigator, debugger, or explainer. Beginners may build one strong scene; advanced students may add loops, agent movement, signage, or an audience-centered accessibility feature.

Lesson 2 Plan

Duration: 60 minutes

Lesson title: Building, Coding, and Debugging the Meaningful Place

Student-friendly target: I can build and revise my Minecraft place so that at least one coded or repeated feature helps communicate meaning.

Joy + identity bridge: Lesson 2 keeps coding joyful by treating debugging as experimentation rather than failure. Students revise features so their worlds become clearer, more welcoming, or more meaningful to an audience. This connects technical skill to identity because the code is not isolated from the story of the place; it helps the place communicate belonging, care, memory, or future possibility.

Objectives:

1. Students will build at least two meaningful features from their plan.
2. Students will use a sequence, loop, repeated structure, or agent-based action to create/improve the world.
3. Students will debug one issue by naming the problem, testing a change, and recording what changed.

Mins	Activity	Purpose
0–8	Revisit planning sheets and choose one “must-build” feature and one “try-to-code” feature.	Focuses students before building.
8–15	Teacher models a short debug cycle: predict, run/test, notice, revise.	Makes debugging visible and normal.
15–42	Students build and code with partner roles. Teacher conferences using the debug log.	Develops skills and persistence.
42–52	Screenshot checkpoint: students capture or sketch one feature and label what it means.	Connects product to identity and audience.
52–60	Quick share: one success, one bug, one next step.	Builds community expertise.

Assessment evidence: debug log, teacher conference note, screenshot/sketch checkpoint, and oral explanation. Students can also submit a brief before/after note showing how one design choice changed during debugging. The strongest evidence is the connection between the technical revision and the meaning students want to communicate.

Teacher look-fors: During conferences, the teacher listens for students naming what they tried or changed, and how the feature supports meaning. That evidence can be small: a revised pathway, clearer sign, repeated pattern, debug note, or screenshot label that connects a technical decision to identity, community, or audience.

Lesson 3 Plan

Duration: 60 minutes

Lesson title: Sharing Worlds, Reading Design, and Reflecting on Representation

Student-friendly target: I can present my Minecraft place, explain how it carries identity or community knowledge, and respond respectfully to other students' designs.

Objectives:

1. Students will explain at least two design choices and how they carry meaning.
2. Students will give peer feedback using notice/wonder language.
3. Students will reflect on whose stories are visible and how designers can make worlds more welcoming.

Mins	Activity	Purpose
0–8	Students prepare a two-sentence explanation and choose one screenshot or live view to show.	Supports concise presentation.
8–35	Gallery walk or paired screen-share: students explain their worlds and collect one notice/wonder comment.	Centers oral, visual, and digital literacy.
35–48	Whole-group critical question: “Whose stories do digital worlds usually show, and whose could be missing?”	Makes criticality explicit.
48–56	Students revise one sentence, sign, pathway, or accessibility feature based on feedback.	Turns feedback into action.
56–60	Exit reflection: “My world shows...” “A designer can make digital spaces more welcoming by...”	Captures final evidence.

Assessment evidence: final screenshot/live demonstration, peer feedback note, short presentation, and exit reflection. Students can also submit a brief revision note showing how feedback changed one sentence, sign, pathway, or accessibility feature. The strongest evidence is the connection between the final design choice and how the student wants the world to be read by others.

Teacher look-fors: During sharing, the teacher listens for students explaining what their design shows, why specific choices matter, and how they responded to peer feedback. That evidence can be small: a clearer explanation, revised sign, added pathway, accessibility feature, peer notice/wonder note, or reflection that connects the world to identity, community, representation, or welcome.

Assessment and Evidence

Evidence collected across the unit: planning sheet, debug log, screenshots or sketches, teacher conference notes, peer feedback, short presentation, and exit reflection.

Criterion	Meeting	Developing
Meaningful design	World includes features connected to identity, community, belonging, or future possibility.	World has features, but the meaning needs clearer explanation.
Computational thinking	Student decomposes, sequences, tests, debugs, or uses a repeated/coded feature.	Student builds but needs support naming steps or debugging.
Multimodal literacy	Student uses building, visuals, code/planning, oral explanation, and written labels/reflection.	Student uses some modes but relies on one form of expression.
Critical awareness	Student can name whose stories are represented and how a design can welcome an audience.	Student notices audience or representation with prompting.

Sample and Appendix Tools

Teacher model sample: a small community garden world with an entrance path, welcome sign, gathering circle, repeated fence pattern, and one coded/agent-assisted repeated feature. The model should be meaningful but not perfect so students see revision as normal.

Planning card	Debug log	Share card
• My place is: • This place matters because: • Three important features are: • My first 3–5 build/code steps are:	• I expected: • What happened: • I changed: • I learned:	• My world shows: • One design choice I made was: • One way this world welcomes people is: • One question I still have is:

Reflection

This mini-unit shows how digital literacy can become a space for belonging. Joy is built into choice, creation, debugging, peer support, and visible progress. Identity is built into meaningful place-making and student control over what is shared. The lesson also protects privacy, which matters because identity-centered teaching should allow students to communicate meaning without forcing them to disclose trauma, culture, family history, or community knowledge on demand.

Muhammad's identity pursuit helps frame the work as self-definition. Moje, Luke, Davies, and Street help ask what identity the task makes possible. Skerrett supports the idea that students can be repositioned as capable contributors. Moje's disciplinary literacy frame helps name coding, debugging, design, and explanation as literacy practices.

For my own teaching, the most important design decision is that students are assessed on how intentionally they communicate meaning. That distinction matters for neurodivergent, culturally connected, multilingual, and mixed-identity students because it keeps literacy tied to agency. In this unit, joy is not separate from rigor: students still decompose, debug, revise, and explain, but those skills serve a world they care about.

Appendix: Working-Memory-Friendly Lesson Flow Figures

Design note: These visual figures are meant to support teachers and students who benefit from seeing the whole learning path at once. Each figure separates the build sequence, teacher support, and evidence so the lesson can be followed without holding every step in working memory.

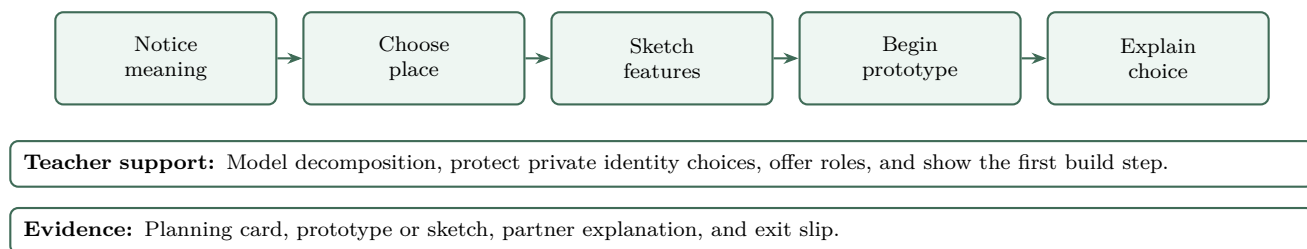


Figure 1: Lesson 1 visual flow: Plan a Meaningful Place.

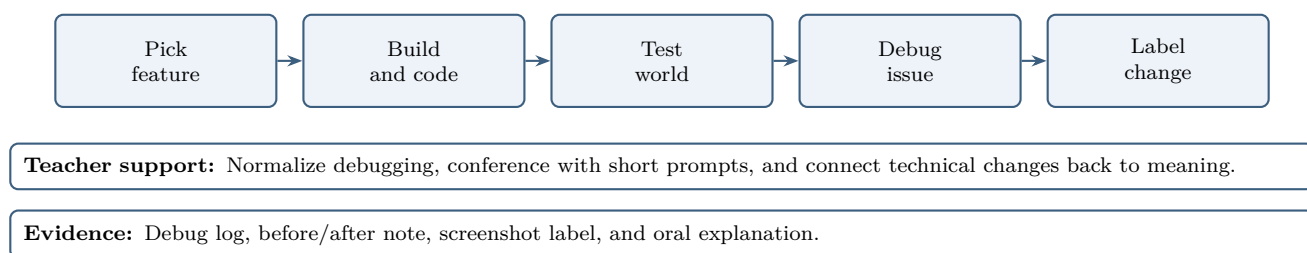


Figure 2: Lesson 2 visual flow: Build, Code, and Debug.

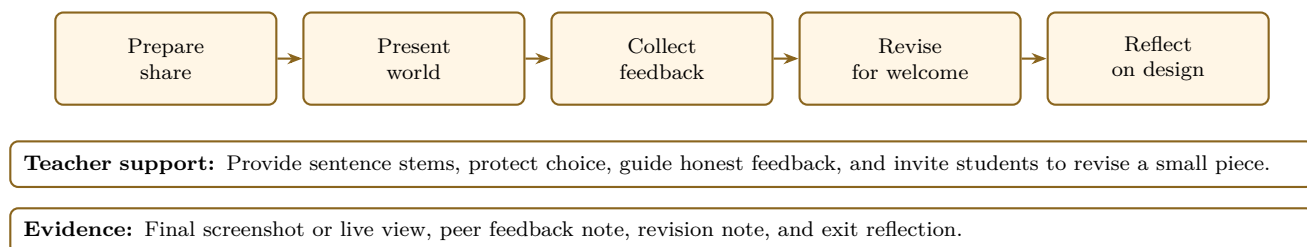


Figure 3: Lesson 3 visual flow: Share, Revise, and Reflect.